

1. Administrative & Study Identification

Full Study Title: A Phase 2 Basket Study of Tucatinib in Combination with Trastuzumab in Subjects with Previously Treated, Locally Advanced Unresectable or Metastatic Solid Tumors Driven by HER2 Alterations **Short Title/Acronym:** SGNTUC-019 **Protocol Number:** SGNTUC-019 **NCT Number:** NCT04579380 **Sponsor Name:** Pfizer **Investigational Product:** Tucatinib and Trastuzumab (Trade names for Trastuzumab include: Herceptin, Enhertu, Ontruzant, Trazimera, Herzuma, Phesgo, Kanjinti, Herceptin Hylecta, Ogivri, KADCYLA). Patients with HR+ HER2-mutated breast cancer also receive fulvestrant. **Phase of Development:** Phase II **Trial Status:** Active, Not Recruiting **Medical Monitor:** Clinical Operations Lead / Medical Monitor, Pfizer

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the confirmed objective response rate (cORR) of tucatinib in combination with trastuzumab in patients with previously treated, locally advanced unresectable or metastatic solid tumors driven by HER2 alterations. **Secondary Objectives:** To evaluate the disease control rate (DCR), duration of response (DOR), progression-free survival (PFS), overall survival (OS), and the safety and tolerability of the treatment regimen.

2.2 Background & Rationale Human epidermal growth factor receptor 2 (HER2) signaling promotes cell growth and is overexpressed or mutated in several tumor types. While HER2-targeted therapies have been highly effective and approved for conditions like HER2-positive breast and gastric cancers, their efficacy across a broader spectrum of solid tumors with HER2 amplification or HER2 mutations (such as biliary tract cancer, cervical cancer, and non-squamous NSCLC) has not been widely investigated. This basket study is designed to evaluate whether vertical receptor inhibition using tucatinib combined with trastuzumab provides clinically meaningful antitumor activity in these previously treated, rare patient populations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Basket Study)
Control Type: Single-arm, multi-cohort **Blinding:** Open-label
Randomization: N/A

3.2 Planned Enrollment & Duration Target **NS:** 217 participants evaluated across multiple cohorts: * 5 tumor-specific cohorts with HER2 overexpression/amplification (cervical cancer, uterine cancer, biliary tract cancer, urothelial cancer, and non-squamous

non-small cell lung cancer [NSCLC]). * 2 tumor-specific cohorts with HER2 mutations (non-squamous NSCLC and breast cancer). * 2 cohorts for all other HER2 amplified/overexpressed solid tumor types (except breast, gastric or gastroesophageal junction adenocarcinoma [GEC], and colorectal cancer [CRC]) or HER2-mutated solid tumor types. **Number of Sites:** International, Multicenter **Estimated Study Start:** 01-Oct-2020 **Current Status:** Active, Not Recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of locally-advanced unresectable or metastatic solid tumor with HER2 alterations (including primary brain tumors). 2. Disease demonstrating HER2 alterations (overexpression/amplification or activating mutations) determined by local or central testing in a CLIA- or ISO-accredited laboratory from fresh/archival tissue or blood. 3. Disease progression during or after ≥ 1 prior line of systemic therapy for locally-advanced unresectable or metastatic disease (or progression during/after standard treatment, or for which no standard treatment is available, for non-squamous NSCLC). 4. Measurable disease per RECIST v1.1 criteria according to investigator assessment. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1.

4.2 Exclusion Criteria 1. Participants with breast cancer, gastric or gastroesophageal junction adenocarcinoma [GEC], or colorectal cancer (CRC) whose disease shows HER2 amplification/overexpression. 2. Previous treatment with HER2-directed therapy (Exceptions: participants with uterine serous carcinoma or HER2-mutated gastric/GEC without HER2-overexpression/amplification may have received prior trastuzumab). 3. Known hypersensitivity to any component of the drug formulation of tucatinib or trastuzumab (drug substance, excipients, murine proteins), or fulvestrant in participants with HR+ HER2-mutated breast cancer. 4. History of exposure to a 360 mg/m^2 doxorubicin-equivalent or $>720 \text{ mg/m}^2$ epirubicin-equivalent cumulative dose of anthracyclines. 5. Treatment with any systemic anti-cancer therapy, radiation therapy, major surgery, or experimental agent within ≤ 3 weeks of first dose of study treatment, or current participation in another interventional clinical trial.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Tucatinib 300 mg administered twice per day. Trastuzumab administered at 8 mg/kg on Day 1 of Cycle 1, followed by 6 mg/kg every 21 days beginning Day 1 of Cycle 2. Patients with HR+ HER2-mutated breast cancer also receive fulvestrant. **Route of Administration:** Tucatinib (Oral); Trastuzumab (Intravenous). **Duration of Treatment:** 21-day cycles, continuing until disease progression or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care; Fulvestrant is permitted and required for the HR+ HER2-mutated breast cancer cohort. **Prohibited:** Concurrent experimental agents and other targeted anti-cancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, HER2 status local/central testing, Baseline tumor imaging
Baseline	Cycle 1, Day 1	First Dose of Tucatinib + Trastuzumab (8 mg/kg loading dose)
Interim	Every 21 Days	Cycle 2+ infusions (Trastuzumab 6 mg/kg), Tucatinib daily dosing, Safety Labs, RECIST v1.1 Tumor Assessments
End of Study	Follow-Up	Final safety assessment, survival tracking (evaluated up to ~2 years)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Confirmed Objective Response Rate (cORR) as assessed by the investigator. Defined as the percentage of participants with a best overall response of complete response (CR) or partial response (PR) according to RECIST v1.1 criteria, and considered confirmed when the subsequent response is at least 4 weeks after the initial response. *Measurement definitions per RECIST v1.1:* * **CR:** Disappearance of all target lesions; any pathological lymph nodes must have reduction in short axis to <10 mm. * **PR:** $\geq 30\%$ decrease in the sum of diameters (SOD) of target lesions. * **Disease Progression (PD):** $\geq 20\%$ relative increase in SOD of target lesions.

7.2 Secondary Endpoints 1. Confirmed Disease Control Rate (DCR) as Assessed by Investigator 2. Duration of Response (DOR) as Assessed by Investigator 3. Progression-Free Survival (PFS) as Assessed by Investigator 4. Overall Survival (OS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for Treatment-Emergent Adverse Events (TEAEs) such as diarrhea, pyrexia, nausea, and cholangitis. **Serious Adverse Events (SAEs):** Standard reporting timelines within 24 hours to the sponsor and Institutional Review Boards. **Data Monitoring Committee (DMC):** Internal safety review committee established by the sponsor (Pfizer).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy analysis is performed on patients with evaluable responses; Safety Set includes all patients receiving at least one dose of study medication. **Sample Size**

Justification: Multi-cohort basket design powered individually per cohort to evaluate a clinically meaningful cORR over historical standard-of-care baseline. **Handling of Missing Data:** Standard oncology imputation; time-to-event endpoints (PFS, OS, DOR) estimated using the Kaplan-Meier method.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and centralized monitoring to ensure accurate RECIST assessment and safety reporting. **Audit Trail:** Electronic Case Report Form (eCRF) utilizing verified source data from participating international oncology centers.

11. Study Identifiers & Governance

Primary ID: SGNTUC-019 **Registry Number:** NCT04579380 **Posting ID:** 84171278297966800 **Sponsor/Lead Organization:** Pfizer **Study Title:** Basket Study of Tucatinib and Trastuzumab in Solid Tumors With HER2 Alterations **Official Regulatory Title:** A Phase 2 Basket Study of Tucatinib in Combination with Trastuzumab in Subjects with Previously Treated, Locally Advanced Unresectable or Metastatic Solid Tumors Driven by HER2 Alterations

12. Clinical Framework (Oncology Specific)

Preferred UMLS Name: HER2 Receptor Positive Cancer **Primary Condition / Disease Areas:** Uterine Neoplasms, Uterine Cervical Neoplasms, Biliary Tract Neoplasms (Biliary tract structure), Urologic Neoplasms, Carcinoma, Non-Small-Cell Lung Cancer, HER2 Mutations Breast Neoplasms **Diagnosis Threshold:** Confirmed HER2 overexpression/amplification or activating HER2 mutation **Medical Methodology:** Targeted vertical HER2 inhibition **Optimization Target:** Objective response and durable disease control

13. Study Procedures & Methodology

Management Pathway: Standard oncology infusion and oral targeted therapy pathway **Education Protocol (HCP):** Site initiation visits and RECIST v1.1 training **Education Protocol (Patient):** Informed consent process detailing the experimental nature of the regimen and oral medication adherence **Quality Audit Mechanism:** Independent/Centralized radiological review (if applicable) and standard clinical auditing

14. Observation & Follow-Up Schedule

Total Duration: Follow-up for up to 2 years **Onsite Visit Cadence:** Every 21 days corresponding with Trastuzumab infusion cycles **Remote Monitoring Frequency:** Between treatment cycles as necessary for AE follow-up **Checklist Review Interval:** Imaging

assessments per protocol specified intervals (e.g., every 6-8 weeks)

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					